

WEBTECH LAB SHEET-4

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SLOT: L37+38

2. Use MongoDB to implement the following DB operations for a Zoo

1. Create a database called 'animal' and write a MongoDB query to select database as 'animal'.
2. Write a MongoDB query to display all the databases.
3. Create a collection called 'wild_animals'.(use capping) and Create a collection called 'domestic_animals'.
4. Add 5 wild_animal details to the collection named 'wild_animals'. Each document consists of following fields as animal_name, nature (harm or harmless), favorite_foods (meat, rabbits, deer etc) as array, care_taker_name, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.
5. Add 5 domestic-animal details to the collection named 'domestic_animals'. Each document consists of following fields as animal_name, gender (male or female), favorite_foods (meat, rabbits, deer etc) as array, animal_petname, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.
6. Write a MongoDB query to display all documents available in wild_animals and domestic_animals.
7. Write a MongoDB query to display only animal name and expenses in all the collection of the database
8. Write a MongoDB query to display domestic_animals whose life is a particular year
9. Write a MongoDB query to display wild_animals available under a particular care_taker
10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.

```

> use animal
< switched to db animal
> db.wild_animals.find()
< {
  _id: ObjectId('6913fa2ffeb4984e49142c0c'),
  animal_name: 'Bengal Tiger',
  nature: 'harm',
  favorite_foods: [
    'meat',
    'deer'
  ],
  care_taker_name: 'Alfred',
  life_span: 15,
  timestamp: 2020-03-15T00:00:00.000Z,
  expenses: 50000
}
{
  _id: ObjectId('6913fa2ffeb4984e49142c0d'),
  animal_name: 'African Lion',
  nature: 'harm',
  favorite_foods: [
    'meat',
    'rabbits'
  ],
  care_taker_name: 'Sree',

```

```

{
  _id: ObjectId('6913fa2ffeb4984e49142c0e'),
  animal_name: 'Indian Elephant',
  nature: 'harmless',
  favorite_foods: [
    'fruits',
    'sugarcane',
    'vegetables'
  ],
  care_taker_name: 'Ramesh Singh',
  life_span: 60,
  timestamp: 2015-01-10T00:00:00.000Z,
  expenses: 75000
}
{
  _id: ObjectId('6913fa2ffeb4984e49142c0f'),
  animal_name: 'Red Panda',
  nature: 'harmless',
  favorite_foods: [
    'bamboo',
    'fruits'
  ],
  care_taker_name: 'Aasamaan',
  life_span: 8,
  timestamp: 2021-05-18T00:00:00.000Z,

```

```

> db.domestic_animals.find()
< {
  _id: ObjectId('6913fa73f530d2918d1ebe77'),
  animal_name: 'Golden Retriever',
  gender: 'male',
  favorite_foods: [
    'meat',
    'dog food'
  ],
  animal_petname: 'Biscuit',
  life_span: 12,
  timestamp: 2021-06-15T00:00:00.000Z,
  expenses: 15000
}
{
  _id: ObjectId('6913fa73f530d2918d1ebe78'),
  animal_name: 'Persian Cat',
  gender: 'female',
  favorite_foods: [
    'fish',
    'meat'
  ],
  animal_petname: 'Whiskey',
  life_span: 15,
  timestamp: 2022-02-20T00:00:00.000Z
}

```

```

  life_span: 8,
  timestamp: 2023-01-10T00:00:00.000Z,
  expenses: 8000
}
{
  _id: ObjectId('6913fa73f530d2918d1ebe7b'),
  animal_name: 'Parrot',
  gender: 'male',
  favorite_foods: [
    'seeds',
    'fruits'
  ],
  animal_petname: 'Arrays',
  life_span: 20,
  timestamp: 2019-08-05T00:00:00.000Z,
  expenses: 10000
}
db.wild_animals.find({}, {animal_name:1, expenses:1, _id:0})
{
  animal_name: 'Bengal Tiger',
  expenses: 50000
}
{
  animal_name: 'African Lion',
  expenses: 48000
}

```

```

}
> db.wild_animals.find({}, {animal_name:1, expenses:1, _id:0})
< {
  animal_name: 'Bengal Tiger',
  expenses: 50000
}
{
  animal_name: 'African Lion',
  expenses: 48000
}
{
  animal_name: 'Indian Elephant',
  expenses: 75000
}
{
  animal_name: 'Red Panda',
  expenses: 25000
}
{
  animal_name: 'Asiatic Black Bear',
  expenses: 45000
}
> db.domestic_animals.find({}, {animal_name:1, expenses:1, _id:0})
< {
  animal_name: 'Golden Retriever',
  expenses: 15000
}

```

```

}
> db.domestic_animals.find({}, {animal_name:1, expenses:1, _id:0})
< {
  animal_name: 'Golden Retriever',
  expenses: 15000
}
{
  animal_name: 'Persian Cat',
  expenses: 12000
}
{
  animal_name: 'Labrador',
  expenses: 18000
}
{
  animal_name: 'Rabbit',
  expenses: 8000
}
{
  animal_name: 'Parrot',
  expenses: 10000
}
> db.domestic_animals.find({life_span:12})
< {
  _id: ObjectId('6913fa73f530d2918d1ebe77'),

```

```

    _id: ObjectId('6913fa73f530d2918d1ebe77'),
    animal_name: 'Golden Retriever',
    gender: 'male',
    favorite_foods: [
      'meat',
      'dog food'
    ],
    animal_petname: 'Biscuit',
    life_span: 12,
    timestamp: 2021-06-15T00:00:00.000Z,
    expenses: 15000
  }
> db.wild_animals.find({care_taker_name:"Ramesh"})
<
> db.wild_animals.find(
  {life_span:{$gt:5}},
  {animal_name:1,favorite_foods:1,expenses:1,_id:0}
)
< {
  animal_name: 'Bengal Tiger',
  favorite_foods: [
    'meat',
    'deer'
  ],
  expenses: 50000
}

```

```

    animal_name: 'Indian Elephant',
    favorite_foods: [
      'fruits',
      'sugarcane',
      'vegetables'
    ],
    expenses: 75000
  }
  {
    animal_name: 'Red Panda',
    favorite_foods: [
      'bamboo',
      'fruits'
    ],
    expenses: 25000
  }
  {
    animal_name: 'Asiatic Black Bear',
    favorite_foods: [
      'meat',
      'fruits',
      'honey'
    ],
    expenses: 45000
  }
}

```

1. Use MongoDB to implement the following DB operations

1. Create a database called 'vehicles' and write a MongoDB query to select database as "vehicles".
2. Write a MongoDB query to display all the databases.
3. Create a collection called 'two_wheelers'. (use capping) and Create a collection called 'four_wheelers'.
4. Add 5 two-wheeler details to the collection named 'two_wheelers'. Each document consists of following fields as bike_name, model (gear or gearless), category (100cc, 125cc, 150cc, 200cc), colors_available (red, black, blue, sport red etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.
5. Add 5 four-wheeler details to the collection named 'four_wheelers'. Each document consists of following fields as vehicle_name, model (commercial or own), category (car, lorry, bus, mini truck, heavy truck, containers), variants (vdi, xdi, petrol, diesel etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.
6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.
7. Write a MongoDB query to display only vehicle name and price in all the collection of the database
8. Write a MongoDB query to display two_wheelers from a particular company
9. Write a MongoDB query to display four_wheelers available in diesel variants
10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.

```
> use vehicles_Sree
< switched to db vehicles_Sree
> db["two_wheelers"].find()
< {
  _id: ObjectId('6913f87c427675fe5d76ffe1'),
  bike_name: 'Hero Splendor Plus',
  model: 'gearless',
  category: '100cc',
  colors_available: [
    'red',
    'black',
    'blue'
  ],
  manufacturer: 'Hero MotoCorp',
  performance: 7,
  timestamp: 2023-01-15T00:00:00.000Z,
  price: 65000
}
{
  _id: ObjectId('6913f87c427675fe5d76ffe2'),
  bike_name: 'Honda Activa 6G',
  model: 'gearless',
  category: '125cc',
  colors_available: [
    'black',
```

```

    ],
    manufacturer: 'Honda',
    performance: 8,
    timestamp: 2022-08-20T00:00:00.000Z,
    price: 75000
  }
  {
    _id: ObjectId('6913f87c427675fe5d76ffe3'),
    bike_name: 'TVS Apache RTR 160',
    model: 'gear',
    category: '150cc',
    colors_available: [
      'red',
      'black',
      'blue'
    ],
    manufacturer: 'TVS',
    performance: 9,
    timestamp: 2023-03-10T00:00:00.000Z,
    price: 115000
  }
  {
    _id: ObjectId('6913f87c427675fe5d76ffe4'),
    bike_name: 'Bajaj Pulsar NS200',
    model: 'gear',

```

```

    price: 125000
  }
  {
    _id: ObjectId('6913f87c427675fe5d76ffe6'),
    bike_name: 'Benelli',
    model: 'gear',
    category: '450cc',
    colors_available: [
      'black',
      'white'
    ],
    manufacturer: 'Race Car',
    performance: 9,
    timestamp: 2022-11-05T00:00:00.000Z,
    price: 145000
  }
db.four_wheelers.find()
{
  _id: ObjectId('6913f8c47495bf5f2fbabae9'),
  vehicle_name: 'Maruti Swift',
  model: 'own',
  category: 'car',
  variants: [
    'vxi',
    'zxi',

```

```

        'ZXT',
        'petrol'
    ],
    manufacturer: 'Maruti Suzuki',
    performance: 7,
    timestamp: 2023-01-20T00:00:00.000Z,
    price: 600000
}
{
    _id: ObjectId('6913f8c47495bf5f2fbabaea'),
    vehicle_name: 'Tata Ace',
    model: 'commercial',
    category: 'mini truck',
    variants: [
        'diesel'
    ],
    manufacturer: 'Tata Motors',
    performance: 6,
    timestamp: 2022-09-15T00:00:00.000Z,
    price: 450000
}
{
    _id: ObjectId('6913f8c47495bf5f2fbabae'),
    vehicle_name: 'Mahindra Scorpio',
    model: 'own',

```

```

    ],
    manufacturer: 'Mahindra',
    performance: 8,
    timestamp: 2023-04-10T00:00:00.000Z,
    price: 1350000
}
{
    _id: ObjectId('6913f8c47495bf5f2fbabae'),
    vehicle_name: 'Ashok Leyland Dost',
    model: 'commercial',
    category: 'mini truck',
    variants: [
        'diesel'
    ],
    manufacturer: 'Ashok Leyland',
    performance: 7,
    timestamp: 2022-12-05T00:00:00.000Z,
    price: 700000
}
{
    _id: ObjectId('6913f8c47495bf5f2fbabae'),
    vehicle_name: 'Honda City',
    model: 'own',
    category: 'car',
    variants: [

```



```

    price: 1200000
  }
> db.two_wheelers.find({}, {bike_name:1, price:1, _id:0})
< {
  bike_name: 'Hero Splendor Plus',
  price: 65000
}
{
  bike_name: 'Honda Activa 6G',
  price: 75000
}
{
  bike_name: 'TVS Apache RTR 160',
  price: 115000
}
{
  bike_name: 'Bajaj Pulsar NS200',
  price: 145000
}
{
  bike_name: 'Yamaha FZ-S FI',
  price: 125000
}
{
  bike_name: 'Benelli',
  price: 145000
}

```

```

}
> db.four_wheelers.find({}, {vehicle_name:1, price:1, _id:0})
< {
  vehicle_name: 'Maruti Swift',
  price: 600000
}
{
  vehicle_name: 'Tata Ace',
  price: 450000
}
{
  vehicle_name: 'Mahindra Scorpio',
  price: 1350000
}
{
  vehicle_name: 'Ashok Leyland Dost',
  price: 700000
}
{
  vehicle_name: 'Honda City',
  price: 1200000
}
> db.four_wheelers.find({variants:"diesel"})
< {
  id: ObjectId('6913f8c47495bf5f2fbabaea').

```

```

}
> db.four_wheelers.find({variants:"diesel"})
< {
  _id: ObjectId('6913f8c47495bf5f2fbabaea'),
  vehicle_name: 'Tata Ace',
  model: 'commercial',
  category: 'mini truck',
  variants: [
    'diesel'
  ],
  manufacturer: 'Tata Motors',
  performance: 6,
  timestamp: 2022-09-15T00:00:00.000Z,
  price: 450000
}
{
  _id: ObjectId('6913f8c47495bf5f2fbabae'),
  vehicle_name: 'Mahindra Scorpio',
  model: 'own',
  category: 'car',
  variants: [
    'diesel',
    'petrol'
  ],
  manufacturer: 'Mahindra',

```

```

{
  _id: ObjectId('6913f8c47495bf5f2fbabae'),
  vehicle_name: 'Honda City',
  model: 'own',
  category: 'car',
  variants: [
    'petrol',
    'diesel'
  ],
  manufacturer: 'Honda',
  performance: 9,
  timestamp: 2023-03-25T00:00:00.000Z,
  price: 1200000
}
> db.four_wheelers.find(
  {performance:{>5}},
  {vehicle_name:1,category:1,manufacturer:1,_id:0}
)
< {
  vehicle_name: 'Maruti Swift',
  category: 'car',
  manufacturer: 'Maruti Suzuki'
}
{
  vehicle_name: 'Tata Ace',

```

```
,  
< {  
  vehicle_name: 'Maruti Swift',  
  category: 'car',  
  manufacturer: 'Maruti Suzuki'  
}  
{  
  vehicle_name: 'Tata Ace',  
  category: 'mini truck',  
  manufacturer: 'Tata Motors'  
}  
{  
  vehicle_name: 'Mahindra Scorpio',  
  category: 'car',  
  manufacturer: 'Mahindra'  
}  
{  
  vehicle_name: 'Ashok Leyland Dost',  
  category: 'mini truck',  
  manufacturer: 'Ashok Leyland'  
}  
{  
  vehicle_name: 'Honda City',  
  category: 'car',  
  manufacturer: 'Honda'  
}
```